Sahaj Bhadja

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Summary

Software Engineer with 2+ years of experience building **distributed services**, **TypeScript/React applications**, REST APIs, cloud infrastructure, and **reliable AI-enabled workflows** at production scale.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering

Experience

USC Dornsife | *Software Engineer* | *Los Angeles, CA* **September 2025 – Present**
- Tripled concurrent throughput across 1K+ AI jobs by restructuring **fault-tolerant microservices** with explicit state transitions, failure isolation, recoverable background processing, and observable workflows under sustained production load reliably.
- Achieved 99%+ reliability across 5K+ monthly jobs through **incremental data pipelines**, implementing validation, reconciliation, error handling, cloud storage, CI/CD releases, and stable production observability reliably at scale.

Grad8 | *Software Engineer Intern* | *Los Angeles, CA* **June 2025 – August 2025**
- Reduced feedback turnaround by 60% through a **production Python platform** supporting 150+ instructors and 10K+ daily requests with REST APIs, AWS RDS, secure ingestion, and CI/CD workflows.
- Processed 1K+ files per batch using **Python multithreading**, validating heterogeneous inputs while isolating failures and preserving responsive user workflows during concurrent model execution and sustained production traffic reliably.

USC AD Labs | *ML Research Assistant* | *Los Angeles, CA* **January 2025 – May 2025**
- Improved long-horizon trajectory planning by 18% in simulation by building a **Decision Transformer pipeline** over 100K+ autonomous-driving logs with offline reinforcement learning and structured evaluation workflows reliably.
- Improved closed-loop planning stability by 15% through a **spatial-temporal encoder** combining HD-map geometry, ego trajectories, nearby-agent context, multi-head attention, and distance-aware interaction modeling across complex simulation scenarios.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer - (SecureSync) | *India*
- Improved API response times by 30% through **object-oriented microservices**, indexing, caching, pagination, and filtered payloads while debugging integrations across Windows, Linux, and macOS production environments reliably at scale.
- Automated builds and 20+ unit/integration suites through **Jenkins, GitHub, PyTest, and Mockito**, generating failure reports and coverage summaries across Linux, Windows, and macOS with repeatable nightly workflows.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern - (VibrantChem) | *India*
- Synchronized 2M+ records through an eight-service **React and Node.js platform** covering authentication, resources, customers, billing, leads, permissions, and notifications with WebSockets and Azure SQL across locations and workflows.
- Reduced resource wastage by 42% through **real-time data synchronization**, implementing JWT access controls, WebSocket updates, automated tests, Azure monitoring, and Docker/Jenkins deployments across eight modular product service domains.

Skills

Languages: Java, Python, TypeScript, JavaScript, C++, SQL, Bash
Frontend & Backend: React, Node.js, FastAPI, Flask, Express.js, REST APIs, WebSockets, Microservices, Background Workers
Distributed Systems: Concurrency, Multithreading, Fault Tolerance, Failure Isolation, State Management, Caching, Data Pipelines, Event-Driven Processing
Cloud & Data: AWS, Azure, Kubernetes, Docker, PostgreSQL, MySQL, MongoDB, Redis, AWS RDS, AWS S3
AI & Quality: LLMs, Transformers, PyTorch, Model Evaluation, RAG, Unit Testing, Integration Testing, PyTest, Mockito, Git, GitHub, Jenkins, CI/CD, Debugging, Observability

Projects

Collaborative Work Orchestration Service - Design Concept
- Designed a **work-orchestration service** concept coordinating tasks, dependencies, approvals, retries, notifications, and audit history through Java services, TypeScript interfaces, REST APIs, event-driven workers, and tenant-isolated storage reliably.

AI-Assisted Change Verification - Design Concept
- Designed an **AI-assisted change-verification** concept that scopes requested edits, reviews generated diffs, executes targeted tests, flags unsafe changes, records evaluation evidence, and requires accountable engineer approval before release.

Big-Brain | Live
- Built a **retrieval-augmented generation platform** using LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP, and REST APIs, sustaining 95% query accuracy while enabling real-time document analysis reliably for 200+ users.